Anesthesia for Disorders of Blood MCQ – Chapter 26

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 26)

50 Questions • 30 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. What is the most common type of anemia encountered in clinical practice in India? **A.** Aplastic anemia **B.** Microcytic hypochromic anemia due to iron deficiency **C.** Hemolytic anemia **D.** Megaloblastic anemia	**Ans: B**
Q2. What is the minimum hemoglobin level considered optimal to accept a patient for elective surgery? **A.** 12 g/dL **B.** 6 g/dL **C.** 10 g/dL **D.** 8 g/dL	**Ans: D**
Q3. What is the formula for calculating maximum allowable blood loss? **A.** (Patient Hct - Desired Hct) / Patient Hct x Blood volume **B.** (Desired Hct - Patient Hct) / Patient Hct x Blood volume **C.** Patient Hct / Desired Hct x Blood volume **D.** (Patient Hct - Desired Hct) / Desired Hct x Blood volume	**Ans: A**
Q4. What is the estimated blood volume used in the calculation for allowable blood loss? **A.** 80-90 mL/kg **B.** 100-110 mL/kg **C.** 60-70 mL/kg **D.** 50-60 mL/kg	**Ans: A**
Q5. What is the current day approach towards minimum use of blood products called? **A.** Restrictive approach to transfusion **B.** Liberal transfusion policy **C.** Aggressive resuscitation **D.** Conservative approach	**Ans: A**
Q6. When is blood transfusion generally indicated in a normal patient? **A.** Blood loss > 20% of blood volume or Hb < 8 g% **B.** Blood loss > 30% of blood volume **C.** Blood loss > 5% of blood volume **D.** Blood loss > 10% of blood volume	**Ans: A**
Q7. At what hemoglobin level is transfusion indicated in a patient with ischemic heart disease? **A.** Less than 10 g% **B.** Less than 12 g% **C.** Less than 8 g% **D.** Less than 6 g%	**Ans: A**

Q8. What type of anesthesia should be avoided for vitamin B12 deficiency anemia?

- **A.** Regional anesthesia
- **B.** Monitored anesthesia care
- **C.** General anesthesia
- **D.** Local anesthesia

Ans: A

Q9. Which inhalational agent should be avoided in megaloblastic anemia?

- **A.** Sevoflurane
- **B.** Nitrous oxide
- **C.** Isoflurane
- **D.** Desflurane

Ans: B

Q10. What effect does alkalosis have on the oxygen dissociation curve?

- **A.** No effect
- **B.** Flattens the curve
- **C.** Shifts it to the right
- **D.** Shifts it to the left

Ans: D

Q11. How is the uptake of inhalational agents affected in anemia patients?

- **A.** No change
- **B.** Increased uptake offset by increased cardiac output
- **C.** Significantly increased uptake
- **D.** Decreased uptake

Ans: B

Q12. What is the risk status of patients having sickle cell trait?

- **A.** Same as sickle cell disease
- **B.** Moderate risk
- **C.** No increase in morbidity and mortality
- **D.** High risk of complications

Ans: C

Q13. Which patients with sickle cell disease should NOT be taken for elective surgery?

- **A.** Patients on hydroxyurea
- **B.** Patients in acute crisis phase
- **C.** Patients with HbS < 30%
- **D.** Patients with normal hematocrit

Ans: B

Q14. What is the target hematocrit for red cell transfusion in sickle cell patients?

- **A.** Hematocrit > 40
- **B.** Hematocrit > 50
- **C.** Hematocrit > 30 (Hemoglobin > 10 g%)
- **D.** Hematocrit > 20

Ans: C

Q15. What is the most important perioperative goal in sickle cell patients?

- **A.** Maintain high hemoglobin
- **B.** Avoid factors that precipitate sickling
- **C.** Maintain hypothermia
- **D.** Use only general anesthesia

Ans: B

Q16. Which factors can precipitate sickling in sickle cell disease? **A.** Only hypoxia **B.** Alkalosis and hyperhydration **C.** Hyperoxia and warmth **D.** Hypoxia, acidosis, dehydration, hypothermia, stress and pain	**Ans: D**

Q17. How should dehydration be managed preoperatively in sickle cell patients? **A.** IV fluids should be started in preoperative period **B.** Fluid restriction **C.** No fluids needed **D.** Oral fluids only	**Ans: A**

Q18. What premedication is recommended for sickle cell patients to avoid stress and anxiety? **A.** Benzodiazepines **B.** Antihistamines **C.** Opioids **D.** Anticholinergics	**Ans: A**

Q19. Is intravenous regional anesthesia (Bier's block) safe in sickle cell patients? **A.** Only without tourniquet **B.** Only with short procedures **C.** No, it does not hold true in current day practice due to tourniquet use **D.** Yes, it is the preferred technique	**Ans: C**

Q20. Can tourniquet be used in sickle cell patients? **A.** Absolutely contraindicated **B.** Can be used safely with minimal possible pressure for minimal period **C.** Can be used without any precautions **D.** Only pneumatic tourniquet allowed	**Ans: B**

Q21. What must be avoided at any cost in sickle cell patients? **A.** Regional anesthesia **B.** Blood transfusion **C.** Inhalational agents **D.** Factors which precipitate sickling	**Ans: D**

Q22. When does acute lung syndrome usually occur postoperatively in sickle cell patients? **A.** Within 24 hours **B.** Immediately after extubation **C.** After 1 week **D.** After 48-72 hours in postoperative period	**Ans: D**

Q23. What is the risk level of patients with thalassemia minor? **A.** Not high risk **B.** Very high risk **C.** Moderate risk **D.** High risk	**Ans: A**

Q24. What can skeletal abnormalities of maxilla in thalassemia cause? **A.** Laryngospasm **B.** Bronchospasm **C.** Easy intubation **D.** Difficult intubation	**Ans: D**

Q25. What organ involvement should be assessed in thalassemia major patients? **A.** Only spleen **B.** Only liver **C.** Hepatosplenomegaly, skeletal involvement, CHF, and consequences of iron overload including cirrhosis **D.** Only cardiac	**Ans: C**

Q26. What is the target hematocrit for polycythemia patients before elective surgery? **A.** Less than 45 **B.** Less than 50 **C.** Less than 55 **D.** Less than 35	**Ans: A**

Q27. What additional bleeding risk exists in polycythemia patients? **A.** Vitamin K deficiency **B.** Increased fibrinolysis **C.** Decreased von Willebrand factor **D.** DIC	**Ans: C**

Q28. What is the recommended platelet count for considering surgery in thrombocytopenia? **A.** > 50,000/mm3 **B.** > 150,000/mm3 **C.** > 100,000/mm3 **D.** > 25,000/mm3	**Ans: A**

Q29. At what platelet count can neurosurgical patients be considered for surgery? **A.** > 150,000 **B.** > 100,000 **C.** > 50,000 **D.** > 30,000	**Ans: B**

Q30. Which patients need platelet transfusion before surgery regardless of platelet count? **A.** Asymptomatic patients **B.** Symptomatic patients with spontaneous bleeding **C.** All surgical patients **D.** Only cardiac patients	**Ans: B**
Q31. Above what platelet count can central neuraxial blocks be safely given? **A.** > 100,000/cubic mm **B.** > 50,000/cubic mm **C.** > 30,000/cubic mm **D.** > 80,000/cubic mm	**Ans: D**
Q32. What is the relative contraindication range for platelet count regarding central neuraxial blocks? **A.** < 30,000 **B.** < 50,000 **C.** Between 80,000 and 100,000 **D.** Between 50,000 and 80,000/cubic mm	**Ans: D**
Q33. What should the deficit coagulation factor be increased to before considering a patient for surgery? **A.** At least 25% of normal **B.** At least 50% of normal **C.** At least 75% of normal **D.** At least 100% of normal	**Ans: B**
Q34. How can coagulation factors be replenished preoperatively? **A.** Fresh frozen plasma, cryoprecipitate or recombinant factor **B.** Vitamin K only **C.** Platelet transfusion **D.** Whole blood only	**Ans: A**
Q35. What type of anesthesia is contraindicated in coagulation deficiency disorders? **A.** Regional anesthesia **B.** Local anesthesia **C.** General anesthesia **D.** IV sedation	**Ans: A**
Q36. Why should intramuscular route be avoided in coagulation disorders? **A.** Risk of hematoma formation **B.** Pain **C.** Infection risk **D.** Poor absorption	**Ans: A**

Q37. Who should perform intubation in patients with massive bleeding in oral cavity from coagulation disorders? **A.** Expert hand with minimum trauma **B.** Any anesthetist **C.** Surgeon **D.** Nurse	**Ans: A**
Q38. For how long should antithrombin levels be maintained > 80% postoperatively in hypercoagulable disorders? **A.** 2 days **B.** 10 days **C.** At least 5 days **D.** 1 day	**Ans: C**
Q39. Why are central neuraxial blocks preferred in hypercoagulable disorders? **A.** They decrease incidence of thromboembolic complications **B.** Better pain control **C.** Shorter duration **D.** Easier to perform	**Ans: A**
Q40. When should DVT prophylaxis be started in hypercoagulable disorders? **A.** Only if symptomatic **B.** As early as possible **C.** After 48 hours **D.** Postoperatively only	**Ans: B**
Q41. What does the anesthetic management of G6PD deficiency patients include? **A.** Only regional anesthesia **B.** Assessing severity of anemia and avoiding drugs that trigger hemolysis **C.** Only monitoring hemoglobin **D.** Mandatory blood transfusion	**Ans: B**
Q42. Which benzodiazepines are safe in G6PD deficiency? **A.** Only lorazepam **B.** Diazepam and midazolam **C.** None are safe **D.** Only alprazolam	**Ans: B**
Q43. Which inhalational agents can trigger hemolysis in G6PD deficiency? **A.** None of them **B.** Isoflurane and sevoflurane **C.** Nitrous oxide only **D.** Halothane and desflurane	**Ans: B**

Q44. Which local anesthetics should be avoided in G6PD deficiency? **A.** All local anesthetics **B.** Only lidocaine **C.** Only bupivacaine **D.** Prilocaine, benzocaine, lignocaine and nitroglycerine	**Ans: D**
Q45. What is the danger of administering methylene blue to G6PD deficient patients? **A.** It can be life threatening **B.** It causes hypertension **C.** No danger **D.** It causes bradycardia	**Ans: A**
Q46. What conditions can trigger hemolysis in G6PD deficiency besides drugs? **A.** Only infection **B.** Only surgery **C.** Only acidosis **D.** Acidosis, hypothermia, hyperglycemia and infection	**Ans: D**
Q47. Which drugs are absolutely contraindicated in porphyria? **A.** Suxamethonium **B.** Barbiturates (Thiopentone/Methohexitone) **C.** Propofol **D.** Opioids	**Ans: B**
Q48. Which agents are considered safe in porphyria patients? **A.** Propofol, opioids, and suxamethonium **B.** Diazepam and pentazocine **C.** Barbiturates and etomidate **D.** Ketamine and muscle relaxants	**Ans: A**
Q49. Should regional anesthesia be used in porphyria patients with peripheral neuropathy? **A.** Only spinal **B.** Yes, it is preferred **C.** No, it should be avoided **D.** Only epidural	**Ans: C**
Q50. What can fasting, dehydration and infection trigger in porphyria patients? **A.** Acute crises **B.** Cardiac arrest **C.** Seizures only **D.** Hemolysis	**Ans: A**

Answer Key & Explanations

Q1. Answer: B — Microcytic hypochromic anemia due to iron deficiency

The most common anemia encountered in clinical practice in India is microcytic hypochromic due to iron deficiency.

Topic: Anemia

Q2. Answer: D — 8 g/dL

A minimum hemoglobin of 8 g/dL is considered optimal to accept the patient for elective surgery.

Topic: Anemia

Q3. Answer: A — (Patient Hct - Desired Hct) / Patient Hct x Blood volume

Maximum allowable blood loss = (Hematocrit of patient - Desirable hematocrit) / Hematocrit of patient x Blood volume.

Topic: Anemia

Q4. Answer: A — 80-90 mL/kg

Blood volume of 80-90 mL/kg is used in the calculation for maximum allowable blood loss.

Topic: Anemia

Q5. Answer: A — Restrictive approach to transfusion

The approach in present day practice is towards the minimum use of blood products called restrictive approach to transfusion.

Topic: Anemia

Q6. Answer: A — Blood loss > 20% of blood volume or Hb < 8 g%

Blood loss greater than 20% of blood volume or hemoglobin level less than 8 g% in a normal patient indicates transfusion.

Topic: Anemia

Q7. Answer: A — Less than 10 g%

In a patient with major disease like ischemic heart disease, transfusion is indicated when Hb is less than 10 g%.

Topic: Anemia

Q8. Answer: A — Regional anesthesia

Regional anesthesia should be avoided for vitamin B12 deficiency anemia due to possibility of neurological deficits.

Topic: Anemia

Q9. Answer: B — Nitrous oxide

Nitrous oxide can cause megaloblastic anemia therefore should be avoided for vitamin B12 deficiency anemia.

Topic: Anemia

Q10. Answer: D — Shifts it to the left

Alkalosis (especially caused by hyperventilation) shifts the oxygen dissociation curve to the left, increasing the oxygen requirement.

Topic: Anemia

Q11. Answer: B — Increased uptake offset by increased cardiac output

Increase uptake of inhalational agents is offset by increased cardiac output seen in anemia patients.

Topic: Anemia

Q12. Answer: C — No increase in morbidity and mortality

There is no increase in morbidity and mortality in patients having sickle cell trait while patients with sickle cell disease are at high risk.

Topic: Sickle Cell Disease

Q13. Answer: B — Patients in acute crisis phase

Patients in acute crisis phase must not be taken for elective surgery.

Topic: Sickle Cell Disease

Q14. Answer: C — Hematocrit > 30 (Hemoglobin > 10 g%)

The aim is to correct the anemia by giving red cell transfusion with the target hematocrit > 30 (Hemoglobin > 10 g%).

Topic: Sickle Cell Disease

Q15. Answer: B — Avoid factors that precipitate sickling

The most important perioperative goal is to avoid factors which can precipitate sickling: hypoxia, acidosis, dehydration, hypothermia, stress and pain.

Topic: Sickle Cell Disease

Q16. Answer: D — Hypoxia, acidosis, dehydration, hypothermia, stress and pain

Factors that precipitate sickling include hypoxia, acidosis, dehydration, hypothermia, stress and pain.

Topic: Sickle Cell Disease

Q17. Answer: A — IV fluids should be started in preoperative period

To avoid dehydration, intravenous fluids should be started in preoperative period.

Topic: Sickle Cell Disease

Q18. Answer: A — Benzodiazepines

To avoid stress and anxiety these patients must be premedicated with benzodiazepines.

Topic: Sickle Cell Disease

Q19. Answer: C — No, it does not hold true in current day practice due to tourniquet use

The old recommendation of giving IV regional anesthesia (Bier's block) to sickle cell patients does not hold true in current day practice due to tourniquet use.

Topic: Sickle Cell Disease

Q20. Answer: B — Can be used safely with minimal possible pressure for minimal period

Per current guidelines, tourniquet can be used safely in sickle cell patients with a little higher complications. It is recommended to use for minimal period with minimal possible pressure.

Topic: Sickle Cell Disease

Q21. Answer: D — Factors which precipitate sickling

The factors which precipitate sickling must be avoided at any cost.

Topic: Sickle Cell Disease

Q22. Answer: D — After 48-72 hours in postoperative period

Acute lung syndrome usually occurs after 48-72 hours in postoperative period.

Topic: Sickle Cell Disease

Q23. Answer: A — Not high risk

Patients with thalassemia minor are not high risk while those with thalassemia major are high risk patients for perioperative morbidity and mortality.

Topic: Thalassemia

Q24. Answer: D — Difficult intubation

Skeletal abnormalities of maxilla (maxillary overgrowth) can make intubation difficult.

Topic: Thalassemia

Q25. Answer: C — Hepatosplenomegaly, skeletal involvement, CHF, and consequences of iron overload including cirrhosis

It is important to assess organ involvement like hepatosplenomegaly, skeletal involvement, congestive heart failure and consequences of iron overload like cirrhosis.

Topic: Thalassemia

Q26. Answer: A — Less than 45

Reduce hematocrit to <45 and platelet count to less than 4 lakhs before taking the patient for elective surgery.

Topic: Polycythemia

Q27. Answer: C — Decreased von Willebrand factor

Patients with polycythemia are at increased risk of bleeding due to decrease von Willebrand factor seen in polycythemia.

Topic: Polycythemia

Q28. Answer: A — > 50,000/mm3

Platelet count of > 50,000/mm3 is recommended before considering for surgery.

Topic: Thrombocytopenia

Q29. Answer: B — > 100,000

Neurosurgical patients cannot be considered for surgery if their platelet count is < 100,000.

Topic: Thrombocytopenia

Q30. Answer: B — Symptomatic patients with spontaneous bleeding

Symptomatic patients (spontaneous bleeding) irrespective of their platelet count need platelet transfusion before surgery.

Topic: Thrombocytopenia

Q31. Answer: D — > 80,000/cubic mm

Patients with platelet count more than 80,000/cubic mm can be safely given central neuraxial blocks.

Topic: Thrombocytopenia

Q32. Answer: D — Between 50,000 and 80,000/cubic mm

Platelet count between 50,000 and 80,000/cubic mm is a relative contraindication while count less than 50,000 is an absolute contraindication.

Topic: Thrombocytopenia

Q33. Answer: B — At least 50% of normal

The deficit factor should be increased to at least 50% of the normal before considering the patient for surgery.

Topic: Coagulation Disorders

Q34. Answer: A — Fresh frozen plasma, cryoprecipitate or recombinant factor

The deficit factor can be increased by administration of fresh frozen plasma, cryoprecipitate or recombinant factor.

Topic: Coagulation Disorders

Q35. Answer: A — Regional anesthesia

Regional anesthesia is contraindicated in coagulation deficiency disorders.

Topic: Coagulation Disorders

Q36. Answer: A — Risk of hematoma formation

Intramuscular route should be avoided in coagulation disorders due to risk of bleeding/hematoma.

Topic: Coagulation Disorders

Q37. Answer: A — Expert hand with minimum trauma

Intubation should be performed by expert hand with minimum trauma, otherwise there can be massive bleeding in oral cavity.

Topic: Coagulation Disorders

Q38. Answer: C — At least 5 days

Antithrombin levels should be > 80% at least for 5 days in postoperative period.

Topic: Hypercoagulable Disorders

Q39. Answer: A — They decrease incidence of thromboembolic complications

Central neuraxial blocks are preferred because they decrease the incidence of thromboembolic complications (due to increase blood in lower limb and pelvic veins).

Topic: Hypercoagulable Disorders

Q40. Answer: B — As early as possible

All prophylaxis against deep venous thrombosis (DVT) should be started as early as possible.

Topic: Hypercoagulable Disorders

Q41. Answer: B — Assessing severity of anemia and avoiding drugs that trigger hemolysis

The anesthetic management includes assessing the severity of anemia and avoiding drugs and conditions which can trigger hemolysis in G6PD deficient individuals.

Topic: G6PD Deficiency

Q42. Answer: B — Diazepam and midazolam

Diazepam; interestingly, midazolam is safe in G6PD deficiency.

Topic: G6PD Deficiency

Q43. Answer: B — Isoflurane and sevoflurane

Isoflurane and sevoflurane can trigger hemolysis in perioperative period in G6PD deficient patients.

Topic: G6PD Deficiency

Q44. Answer: D — Prilocaine, benzocaine, lignocaine and nitroglycerine

Prilocaine, benzocaine, lignocaine and nitroglycerine should be avoided as they can cause methemoglobinemia which can cause significant hemolysis.

Topic: G6PD Deficiency

Q45. Answer: A — It can be life threatening

Administrating methylene blue to these patients can be life threatening.

Topic: G6PD Deficiency

Q46. Answer: D — Acidosis, hypothermia, hyperglycemia and infection

Acidosis, hypothermia, hyperglycemia and infection can trigger hemolysis and must be avoided throughout in these patients.

Topic: G6PD Deficiency

Q47. Answer: B — Barbiturates (Thiopentone/Methohexitone)

Barbiturates (Thiopentone/Methohexitone) are absolutely contraindicated as they induce aminolevulinic acid synthetase and precipitate all kinds of porphyrias except porphyria cutanea tarda and erythropoietic protoporphyrias.

Topic: Porphyria

Q48. Answer: A — Propofol, opioids, and suxamethonium

Propofol, opioids, suxamethonium are considered as safe agents in porphyria.

Topic: Porphyria

Q49. Answer: C — No, it should be avoided

Regional anesthesia should be avoided in the porphyria patients with peripheral neuropathy.

Topic: Porphyria

Q50. Answer: A — Acute crises

Fasting, dehydration and infection can trigger acute crises in porphyria patients.

Topic: Porphyria